tracts namely **ascending tracts** which carry sensory impulses to the brain and descending **tracts** which carry motor impulses from the brain to the spinal nerves at various levels of the spinal cord. The spinal cord shows two enlargements, one in the cervical region and another one in the lumbosacral region. The **cervical enlargement** serves the upper limb and **lumbar enlargement** serves the lower limbs.

10.5 Reflex action and Reflex arc

When dust falls in our eyes, the eyelids close immediately not waiting for our willingness; on touching a hot pan, the hand is withdrawn rapidly. Do you know how this happens?

The spinal cord remains as a connecting functional nervous structure in between the brain and effector organs. But sometimes when a very quick response is needed, the spinal cord can effect motor initiation as the brain and brings about an effect. This rapid action by spinal cord is called reflex action. It is a fast, involuntary, unplanned sequence of actions that occurs in response to a particular stimulus. The nervous elements involved in carrying out the reflex action constitute a reflex arc or in other words the pathway followed by a nerve impulse to produce a reflex action is called a reflex arc (Figure 10.12).

Functional components of a reflex arc

Sensory Receptor - It is a sensory structure that responds to a specific stimulus.

Sensory Neuron - This neuron takes the sensory impulse to the grey (afferent) matter of the spinal cord through the dorsal root of the spinal cord.

Interneurons - One or two interneurons may serve to transmit the impulses from the sensory neuron to the motor neuron.

Motor Neuron - it transmits impulse from CNS to the effector organ.

Effector Organs -It may be a muscle or gland which responds to the impulse received.

There are two types of reflexes. They are

1) **Unconditional reflex** is an inborn reflex for an unconditioned stimulus. It does not need any past experience, knowledge or training to occur; Ex: blinking of an eye when a dust particle about to fall into it, sneezing and coughing due to foreign particle entering the nose or larynx.

2) **Conditioned reflex** is a respone to a stimulus that has been acquired by learning. This does not naturally exists in animals. Only an experience makes it a part of the behaviour. Example: excitement of salivary gland on seeing and smelling a food. The conditioned reflex was first demonstrated by the Russian physiologist **Pavlov** in his classical conditioning experiment in a dog. The cerebral cortex controls the conditioned reflex.

Peripheral Neural System (PNS)

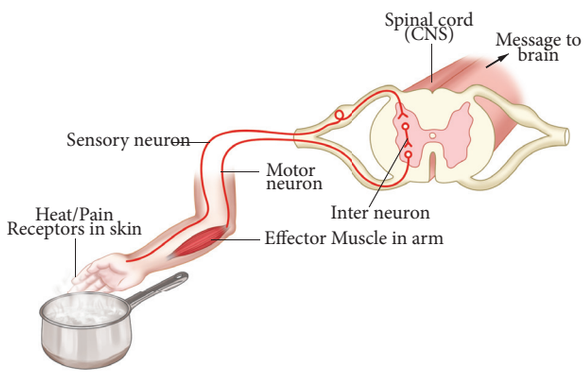


Figure 10.12 Reflex arc

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